

LINEAR ALGEBRA AND CALCULUS

UNIT-1

1). Find the Inverse of the matrix $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$ by Gauss-Jordan Method

2). Determine the rank of the matrix (a) $\begin{pmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \end{pmatrix}$ (b) $\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$

3). Determine the rank of the matrix (a) $\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$ (b) $\begin{bmatrix} 8 & 1 & 3 & 6 \\ 0 & 3 & 2 & 2 \\ -8 & -1 & -3 & 4 \end{bmatrix}$ by reducing into the normal form

4). Test for consistency and solve the following system of equations

$$5x + 3y + 7z = 4, 3x + 26y + 2z = 9, 7x + 2y + 10z = 5$$

5) Find the values of “a” & “b” for which the equations $x - ay + z = 3, x + 2y + 2z = b, x + 5y + 3z = 9$ will have (i) Unique solution (ii) Infinite no. of solutions (iii) No solution

6). Investigate the values of λ and μ so that the equations $2x + 3y + 5z = 9, 7x + 3y - 2z = 8, 2x + 3y + \lambda z = \mu$ have (i) no solution (ii) a unique solution (iii) an infinite number of solutions.

7) Find the values of α and β for which the equations $x + 2y + 3z = 6, x + 3y + 5z = 9, 2x + 5y + \alpha z = \beta$ have (i) Unique solution (ii) Infinite no. of solutions (iii) No solution

8). Solve the following system of equations by Gauss-Elimination method

(a) $x + y + z = 9, 2x - 3y + 4z = 13, 3x + 4y + 5z = 40$

(b) $2x_1 + x_2 + x_3 = 10, 3x_1 + 2x_2 + 3x_3 = 18, x_1 + 4x_2 + 9x_3 = 16$

9). Solve the Following Equations by Gauss-Jacobi Iterative Method

$$10x + y - z = 11.19, x + 10y + z = 28.08, x + y + 10z = 35.61$$

10). Solve the Following Equations by Gauss-Seidel Iterative Method

(a) $17x - 6y - 3z = -3, -4x + 22y + 9z = 3, 2x + 22z = -22$

(b) $10x + y + z = 12, 2x + 10y + z = 13, 2x + 2y + 10z = 14$

11) Solve the equations $x + y + w = 0, y + z = 0, x + y + z + w = 0, x + y + 2z = 0.$

UNIT-2

12). Verify the Cayley – Hamilton theorem for the following matrices also find A^{-1} for

(a) $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$ (b) $\begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$ (c) $\begin{bmatrix} 7 & 2 & -2 \\ -6 & -1 & 2 \\ 6 & 2 & -1 \end{bmatrix}$

13). Verify the Cayley – Hamilton theorem for the Matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$

and find the matrix obtained by $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$

14). Determine the Eigen values and Eigen vectors of the following matrices

(a) $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ (b) $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$

15). Reduce the following Quadratic forms to canonical form by Orthogonal transformation also find the Rank, index, signature & nature

(a) $x^2 + 5y^2 + z^2 + 2xy + 2yz + 6zx$
 (b) $3x^2 + 5y^2 + 3z^2 - 2yz + 2xz - 2xy$
 (c) $3x^2 - 2y^2 - z^2 + 12yz + 8xz - 4xy$

UNIT-3

16). Verify Rolle's Theorem for (a) $f(x) = \log \left(\frac{x^2 + ab}{x(a+b)} \right)$ in (a, b)

(b) $f(x) = \sin x / e^x$ in $(0, \pi)$ (c) $f(x) = x(x+3)e^{-\frac{x}{2}}$ in $[-3, 0]$

(d) $f(x) = (x-a)^m(x-b)^n$ Where m, n are positive integers $[a, b]$.

17). Verify Lagrange's mean value theorem for $f(x) = (x-1)(x-2)(x-3)$ in $(0, 4)$

18). State Lagrange's Mean value theorem and if $f(x) = \sin^{-1}(x)$, $0 < a < b < 1$

use Mean value theorem to Prove that $\frac{b-a}{\sqrt{1-a^2}} < \sin^{-1} b - \sin^{-1} a < \frac{b-a}{\sqrt{1-b^2}}$

19). P.T if $(0 < a < b < 1)$, $\frac{b-a}{1+b^2} < \tan^{-1} b - \tan^{-1} a < \frac{b-a}{1+a^2}$.

Hence S.T $\frac{\pi}{4} + \frac{3}{25} < \tan^{-1} \frac{4}{3} < \frac{\pi}{4} + \frac{1}{6}$

20). Verify Cauchy's mean value theorem for the functions (a) $\sin x$ & $\cos x$ in $(0, \pi)$

(b) Verify Cauchy's mean value theorem for $f(x) = \sqrt{x}$ and $g(x) = \frac{1}{\sqrt{x}}$

21). Using Taylor's theorem, express the polynomial $2x^3 + 7x^2 + x - 6$ in powers of $(x-1)$.

22). Obtain Taylors series expansion of $\sin x$ in powers of $x - \frac{\pi}{4}$.

23). Obtain the Maclaurin's series expansion of e^x

UNIT-4

- 24). If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, show that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = \frac{-9}{(x+y+z)^2}$
- 25). If $z = f(x + ct) + \phi(x - ct)$, prove that $\frac{\partial^2 z}{\partial t^2} = c^2 \frac{\partial^2 z}{\partial x^2}$
- 26). Find the first and second partial derivatives of $z = x^3 + y^3 - 3axy$
- 27). If $u = \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right)$, Prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0$.
- 28). If $u = \log\left(\frac{x^2 + y^2}{x + y}\right)$ Prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 1$
- 29). Expand $x^2y + 3y - 2$ in powers of $(x-1)$ and $(y+2)$
- 30). Expand $e^x \log(1 + y)$ in powers of x and y
- 31). If $u = \sqrt{x^2 + y^2}$ and $v = \tan^{-1}\left(\frac{y}{x}\right)$, find $J\left(\frac{u,v}{x,y}\right)$
- 32). If $u = x^2 - y^2, v = 2xy$ where $x = r \cos \theta, y = r \sin \theta$ then show that $\frac{\partial(u,v)}{\partial(r,\theta)} = 4r^3$
- 33). If $u = x + y + z, uv = y + z, uvw = z$ then show that $\frac{\partial(x,y,z)}{\partial(u,v,w)} = u^2 v$
- 34). In spherical polar coordinates $x = r \sin \theta \cos \phi, y = r \sin \theta \sin \phi, z = r \cos \theta$,
 show that $\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)} = r^2 \sin \theta$.
- 35). Verify whether the following functions are functionally dependent. If so find the relation between them $u = \frac{x+y}{1-xy}, v = \tan^{-1} x + \tan^{-1} y$
- 36). Discuss the maximum and minimum values of $f(x, y) = 1 - x^2 - y^2$
- 37). Examine the extreme values of $x^3 + y^3 + 3axy, a > 0$
- 38). Find the maximum and minimum values of the function
 $f(x, y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$
- 39). A rectangular box open at the top is to have volume of 32 cubic ft. Find the dimensions of the box requiring least material for its construction
- 40). Find the maximum and minimum values of $x + y + z$ subject to $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1$
- 41). Find the maximum & min distance of the point $(3,4,12)$ from the sphere $x^2 + y^2 + z^2 = 4$
- 42). Apply Lagrange's method to find a point on the $3x + 4y + z - 12 = 0$ which is nearest to the origin.
- 43). Evaluate the double integral $\int_0^5 \int_0^{x^2} x(x^2 + y^2) dx dy$
- 44). Evaluate $\int_R xy dx dy$ where R is the region bounded by X -axis and the line $x = 2a$ and $x^2 = 4ay$

45). Evaluate $\int \int r^3 dr d\theta$ over the area bounded by the circles $r = 2 \sin \theta$ and $r = 4 \sin \theta$

46). Evaluate the following double integrals by changing the order of

integration $\int_0^{4a} \int_{\frac{x^2}{4a}}^{2\sqrt{ax}} dy dx$

47). Change the order of integration and evaluate $\int_0^1 \int_{x^2}^{2-x} xy dx dy$

48). Find $\int_0^{\log 2} \int_0^x \int_0^{x+\log y} e^{x+y+z} dx dy dz$

49). Find the value of the double integral $\int_1^2 \int_1^3 xy^2 dx dy$

50). Evaluate the following double integrals by changing to polar co-ordinate

$\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$

SHORT ANSWER QUESTIONS

1. Define rank of a matrix

2. Explain the (a) Echelon form (b) Normal form of a matrix

3. The value of k such that the Rank of $\begin{bmatrix} 1 & -1 & K \\ 0 & 2 & 1 \\ 1 & 0 & 5 \end{bmatrix}$ is 2

4. Find the Rank of $A = \begin{bmatrix} 8 & 1 & 3 \\ 6 & 0 & 3 \\ 2 & 2 & -8 \end{bmatrix}$

5. Test for consistency of the system $x + y + z = 4, x - y + z = 0, 2x + y + z = 5$

6. Define diagonally dominance property.

7. The characteristic equation of the matrix $\begin{bmatrix} 1 & 3 & 5 \\ 0 & 4 & -3 \\ 0 & 0 & 7 \end{bmatrix}$

8. State Cayley-Hamilton Theorem

9. Find the Eigen values of A^{-1}, A^2 for $A = \begin{bmatrix} 1 & 3 & 4 \\ 0 & 4 & 2 \\ 0 & 0 & -3 \end{bmatrix}$

10. If 1,2,4 are Eigen values of the matrix A then find the Eigen values of $3A^2+5A+3I$

11. Write the matrix of the quadratic form $4x^2 - 2y^2 + z^2 - 2xy + 6zx$

12. The quadratic form corresponding to the symmetric matrix $\begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 2 \\ -1 & 2 & 1 \end{bmatrix}$

13. State Rolle's theorem

14. State Lagrange's mean value theorem.

15. State Cauchy's mean value theorem
16. Write Taylors series expansion of $f(x)$ in powers of $(x-a)$
17. Write Maclaurin's series expansion of $f(x)$ in powers of x
18. Explain why mean value theorem does not hold for $f(x) = x^{1/3}$ in $[-1, 1]$.
19. Expand $\sin x$ in powers of x
20. State Euler's theorem.
21. Define saddle point.
22. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ if $z = x^2y - y\sin(xy)$
23. If $x = u(1 + v), y = v(1 + u)$ provethat $J\left(\frac{u,v}{x,y}\right) = 1 + u + v$
24. State Taylor's series of two variables.
25. State Maclaurin's series of two variables
26. If $x = \frac{u^2}{v}, y = \frac{v^2}{u}$ find $\frac{\partial(x,y)}{\partial(u,v)}$.
27. Evaluate $\int_0^2 \int_0^x y dy dx$
28. Find $\int_0^1 \int_1^2 \int_2^3 xyz dx dy dz$
29. Evaluate $\int_0^2 \int_0^x e^{x+y} dx dy$
30. Evaluate $\int_0^1 \int_0^2 \int_0^2 x^2 yz dx dy dz$
31. Evaluate $\int_0^1 \int_y^{\sqrt{y}} x^2 y^2 dx dy$
32. If $x = r\cos\theta, y = r\sin\theta$ then $J\left(\frac{x,y}{u,v}\right)$
33. Evaluate $\int_0^\pi \int_0^a r^3 \sin\theta \cos\theta dr d\theta$
34. Change into polar coordinates $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$
35. Change the order of integration $\int_{-1}^1 \int_0^{\sqrt{1-x^2}} f(x,y) dy dx$